

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	10% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	V.JAGADISH	Reg. No.:	192521352
Section / Batch:		Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze the given questions thoroughly before answering.
- Compare multiple aspects such as methods, results, advantages, and limitations clearly.
- Critically evaluate the strengths, weaknesses, and implications with proper justification.
- Present meaningful insights, interpretations, and well-supported conclusions from the comparisons.

Question Type	Focus Area	Questions
Comparative Analysis	Comparative analysis of Dynamic Programming and Greedy algorithms for optimization problems.	Q1

SECTION A – Comparative Analysis

Q1 String Matching (LCS vs Greedy) 100 marks Given two strings "ABCBDBAB" and "BDCAB", compute the Longest Common Subsequence using Dynamic Programming and compare with a greedy matching attempt. Analyze differences in results and correctness. Evaluate why greedy fails. Generate insights on optimal substructure.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Comparative Analysis					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%				
Comparison & Differentiation	25%				
Critical Evaluation	25%				
Synthesis & Insight Generation	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Algorithm: LCS Using Dynamic Programming

Input: X = "ABCBDBAB" and Y = "BDCAB"

Output: Longest Common Subsequence and its length.

1. **Start**
2. Read the two strings X and Y.
3. Let $m = \text{length}(X)$ and $n = \text{length}(Y)$.
4. Create a DP table $\text{dp}[m+1][n+1]$.
5. Initialize the first row and first column to 0.
6. For $i = 1$ to m :
 - For $j = 1$ to n :
 - If $X[i-1] == Y[j-1]$:
 - Set $\text{dp}[i][j] = \text{dp}[i-1][j-1] + 1$.
 - Otherwise:
 - Set $\text{dp}[i][j] = \max(\text{dp}[i-1][j], \text{dp}[i][j-1])$.
7. The value $\text{dp}[m][n]$ gives the **length of the LCS**.
8. Start from $\text{dp}[m][n]$ and trace backward:
 - If $X[i-1] == Y[j-1]$, include that character in the LCS and move diagonally.
 - Otherwise, move to the neighboring cell with the larger value.
9. Reverse the collected characters to obtain the LCS.
10. Display the LCS and its length.
11. **Stop.**

Result

Input 1: ABCBDAB

Input 2: BDCAB

LCS = BCAB

Length = 4

Time Complexity: $O(m \times n)$

Space Complexity: $O(m \times n)$

QUESTIONS

Q1

String Matching (LCS vs Greedy)
100 marks

0/1 answered

Q1 String Matching (LCS vs Greedy)

100 marks

Given two strings "ABCBDBAB" and "BDCAB", compute the Longest Common Subsequence using Dynamic Programming and compare with a greedy matching attempt. Analyze differences in results and correctness. Evaluate why greedy fails. Generate insights on optimal substructure.

Your Code

```
1 def lcs_dp(a, b):
2     """Return LCS length and one actual LCS using DP."""
3     m, n = len(a), len(b)
4     dp = [[0] * (n + 1) for _ in range(m + 1)]
5
6     # Build DP table
7     for i in range(1, m + 1):
8         for j in range(1, n + 1):
9             if a[i - 1] == b[j - 1]:
10                 dp[i][j] = dp[i - 1][j - 1] + 1
11             else:
12                 dp[i][j] = max(dp[i - 1][j], dp[i][j - 1])
13
14     # Reconstruct one LCS from dp table
15     i, j = m, n
16     lcs_chars = []
17     while i > 0 and j > 0:
18         if a[i - 1] == b[j - 1]:
```

Input

stdin input

Output

```
String 1: ABCBDAB
String 2: BDCAB
-----
DP LCS length: 4
DP LCS: BCAB
```